

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Measure network performance using key metrics with simulation tools
(BL4,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

Question 1

A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.

Answer

The most suitable approach is a hybrid rural connectivity design using a primary 4G/5G fixed-wireless connection, where coverage is available, with a satellite connection as a backup. The primary link should be connected to a business-grade router supporting dual-WAN/failover, QoS, VPN, and traffic monitoring. A managed Gigabit switch should connect office devices, while enterprise Wi-Fi access points should provide wireless access. For security, the branch should use a next-generation firewall, site-to-site IPsec VPN to headquarters, WPA3 for Wi-Fi, strong user authentication, network segmentation, and regular firmware/security updates.

Recommended Design

- Internet connection: Primary 4G/5G fixed wireless; satellite as backup where feasible.
- Router: Dual-WAN business router with automatic failover, QoS, VPN and bandwidth monitoring.
- LAN devices: Managed Gigabit switch and centrally managed wireless access points.

- Security: Next-generation firewall, IPsec site-to-site VPN, WPA3, strong authentication and VLAN-based segmentation.
- Performance: Prioritize video conferencing and critical cloud traffic using QoS; monitor latency, packet loss, throughput and link availability.
- Scalability: Choose equipment with additional Ethernet ports, VLAN support and higher-speed WAN capability so more users and applications can be added later.

Justification

This design balances the constraints because fixed wireless can provide better affordability and lower latency than satellite when a suitable cellular signal exists, while the secondary satellite link improves continuity during primary-link failures. QoS prevents non-critical traffic from consuming the available bandwidth needed for conferencing and cloud services. VPN and firewall controls protect communication with headquarters, while VLANs reduce the impact of compromised devices. The dual-link architecture costs more than a single connection but provides the uninterrupted operation required by the case.

Question 2

A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

Answer

The university should deploy a centrally managed, high-density Wi-Fi design rather than placing access points only for maximum geographic coverage. Access points should be positioned after a wireless site survey, with higher density in classrooms, laboratories and libraries and appropriately spaced coverage in hostels. The 5 GHz band should carry most high-performance client traffic, while 2.4 GHz should be retained for legacy and long-range devices. Where supported, Wi-Fi 6/6E access points should be preferred because they improve efficiency in dense environments.

Recommended Design

- AP placement: Conduct a site survey; place APs in classrooms and high-density areas according to expected client load, not only signal range.
- Channel planning: Use non-overlapping/automatically optimized channels and reduce transmit power where necessary to control co-channel interference.
- Frequency bands: Prefer 5 GHz for most clients; use 2.4 GHz selectively; use 6 GHz where compatible Wi-Fi 6E devices and regulations permit.
- Management: Use a centralized wireless controller/cloud management system for configuration, roaming, monitoring and troubleshooting.

- Authentication: Use WPA3-Enterprise with 802.1X/RADIUS for students and staff, with separate SSIDs or VLANs for students, staff, guests and infrastructure.
- Performance: Apply client/load balancing, QoS and bandwidth policies so a small number of users cannot consume excessive capacity.
- Cost control: Prioritize APs in high-density areas and use wired backbone links; avoid simply increasing AP count because excessive AP density can increase interference.

Justification

The design balances coverage and performance by allocating AP capacity according to user density. Careful channel and transmit-power planning reduces interference, while 5 GHz provides more usable capacity than relying heavily on 2.4 GHz. WPA3-Enterprise and 802.1X provide stronger access control, and VLAN separation limits unnecessary communication between user groups. Centralized management lowers operational effort and makes a limited AP budget more effective.

Question 3

A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

Answer

The financial institution should use a layered architecture based on VLAN segmentation, internal firewalls, strong identity-based authentication and centralized monitoring. Separate VLANs should be created for employees, servers, administrators, guest devices and sensitive systems. A perimeter next-generation firewall should protect the internet edge, while internal firewall/segmentation controls should restrict access between critical zones. Sensitive customer-data systems should be placed in a protected server segment with only the required application flows permitted.

Recommended Secure Architecture

- Segmentation: Use VLANs and Layer-3 access controls for departments; isolate servers, administrators, guests and sensitive customer-data systems.
- Firewall deployment: Use a next-generation firewall at the internet boundary and additional internal segmentation controls for high-value zones.
- Authentication: Use centralized identity management with MFA, role-based access control and least-privilege permissions.

- Intrusion detection: Deploy IDS/IPS at the perimeter and/or critical internal segments, with centralized logging and alert correlation.
- Secure services: Protect online banking and internal applications using TLS, secure management protocols and tightly controlled service ports.
- Performance: Apply firewall rules carefully, use hardware acceleration where available, prioritize business-critical traffic and monitor latency/throughput.
- Compliance and cost: Centralize logs, access policies and monitoring; reuse existing infrastructure where possible and prioritize controls around the most sensitive assets.

Justification

Segmentation limits lateral movement if an endpoint is compromised and makes security policies easier to enforce. MFA and least privilege reduce the risk of stolen credentials, while IDS/IPS provides visibility into suspicious activity. A layered firewall design improves security without requiring every internal device to perform expensive inspection. Centralized management and logging support audit and compliance requirements while controlling operational cost. Performance should be monitored continuously so security inspection does not become a bottleneck.

